

VANI KANJIRANGAT

NLP/AI Researcher (Permanent) | IDSIA – Dalle Molle Institute for Artificial Intelligence Research, Switzerland
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Citations: 800+ | 40+ publications | h-index: 16

RESEARCH SUMMARY

NLP/AI researcher with 8+ years of post-PhD experience (permanent since 2022), specializing in multilingual and dialectal NLP, tokenization and representational fairness in LLMs, uncertainty quantification, biomedical information extraction, and retrieval-augmented generation. Principal Investigator on a Hasler Foundation Young Researcher Grant, Technical NLP Lead on an Armasuisse (Swiss Defense Research) project, and collaborator on applied Financial AI and ESG projects with a leading Swiss banking institution. Recent work on tokenization and dialectal tasks in LLMs received a Senior Area Chair (SAC) Highlight Award at EMNLP 2025 — one of 35 papers selected from 8,172 submissions. Member of the Swiss AI Apertus national initiative (NLP and Tokenization teams), working toward fairer multilingual tokenizers with access to CSCS (Swiss National Supercomputing Centre) HPC infrastructure.

EDUCATION

Ph.D. in Natural Language Processing

Jan 2013 – Aug 2018

Amrita Vishwa Vidyapeetham, India | SERB Junior Research Fellowship, Dept. of Science & Technology, Govt. of India

Thesis: Detection of Extrinsic Text Plagiarism using Natural Language Processing and Intelligent Computing Techniques

M.Tech. in Computer Science

Aug 2010 – May 2012

Cochin University of Science and Technology, India

B.Tech. in Computer Science

Aug 2007 – May 2010

Cochin University of Science and Technology, India

ACADEMIC & RESEARCH POSITIONS

NLP/AI Researcher (Permanent)

Oct 2022 – Present

IDSIA – Dalle Molle Institute for Artificial Intelligence Research, Lugano, Switzerland

- Principal Investigator: Hasler Foundation Young Researcher Grant — biomedical inter-sentential and N-ary relation extraction (2023–present)
- Technical AI-NLP lead, Armasuisse multilingual tokenization project (2023–present)
- Swiss AI Apertus Initiative — Tokenization Team (2025–present) and NLP team (RAG, KGs)
- Collaborator, IDSIA Financial AI Group — RAG, ESG analysis, and KG approaches for banking applications (2024–present)
- Collaborator, IDSIA Imprecise Probability Group — causal reasoning and uncertainty quantification in LLMs
- External supervisor and subject expert for 4 active PhD students across institutions
- Lecturer: NLP & Text Mining, B.Sc. Data Science & AI, SUPSI (2022/23, 2023/24) — continuing 2026/27

Post-Doctoral Researcher

Oct 2018 – Oct 2022

IDSIA – Dalle Molle Institute for Artificial Intelligence Research, Lugano, Switzerland

- Technical AI-NLP lead, Armasuisse dialectal NLP project (2020–2022)
- Biomedical NLP: clinical NER, relation extraction, COVID-19 literature classification
- Innosuisse industry projects: legal knowledge graphs, QA systems, medical device NLP
- Narrative text understanding: temporal embeddings, semantic shift, visual summaries
- Applied Financial AI: interpretability, QA systems

GRANTS & KEY ROLES

Hasler Foundation — Young Researcher Grant (PI)

2023 – Present

Biomedical inter-sentential and N-ary relation extraction using graph neural networks and LLM-based approaches.

Armasuisse Science & Technology (Key Researcher / Technical NLP Lead)

2020 – Present

Multilingual tokenization fairness, dialectal NLP, PEFT fine-tuning across Arabic/Latin/non-Latin settings.

Innosuisse Industrial Collaborations (Researcher)

Legal knowledge graph extraction, QA and recommender systems, medical device NLP.

2018 – 2022

SERB Junior Research Fellowship, Govt. of India

PhD funding for extrinsic text plagiarism detection research.

2013 – 2018

AWARDS & HONOURS

- SAC Highlight Award, EMNLP 2025 (Tokenization, Fairness and Dialectal tasks)
- Best Paper Award — Text2Story@ECIR 2020 (Temporal Embeddings for Narrative Understanding)
- Best Paper Award — ICMMLC 2020, ACM (Temporal Word Embeddings for Narrative Understanding)
- Invited presenter, AMIA Annual Symposium 2022 (N2C2 clinical NLP shared task)

ACADEMIC ACTIVITIES

Teaching

- NLP & Text Mining (theory + lab), B.Sc. Data Science & AI, SUPSI — 2022/23, 2023/24 (planned continuation)
- Software Engineering, B.Tech CS and IT at CET Thrikaripur, Kerala, India — 2012

PhD Supervision (External Supervisor & Subject Expert)

- Small language models and RAG for question generation and answer evaluation — ongoing (2025 – Present)
- Bias mitigation via RL-based prompt selection — ongoing (2025 – Present)
- Scalable AI systems for automated evaluation of short descriptive responses — ongoing (2023 – Present)
- Agricultural NER using Weighted Distributional Semantic Model — completed Aug 2024; patent under review

Shared Task Participation

- NADI 2022 (mono-label) — Arabic dialect classification
- NADI 2024 (multi-label) — Arabic dialect classification
- N2C2 2022 (CMED Track 1) — Clinical NLP; invited presenter at AMIA Annual Symposium
- RealMedNLP (NTCIR-16) — Japanese & English medical NLP
- SemEval-2020 Task 1 — Unsupervised Lexical Semantic Change Detection

Conference Organisation

- SwissText 2022 — Organising & Programme Committee member
- DravidianLangTech@ACL 2026 — Programme Committee member (Speech and Language Technologies for Dravidian Languages)
- LTEDI@ACL 2026 — Programme Committee member (Language Technology for Equality, Diversity and Inclusion)

Invited Talks

- Invited speaker, GenAI and LLMs Bootcamp, Amrita School of Computing and Artificial Intelligence — "From Text to Tokens: Understanding Tokenization in Multilingual LLMs"
- Talk, Cyber Alp Retreat (2025 & 2026), Cyber Defense Campus, Armasuisse — Dialect identification and multilingual tokenization research
- Invited presenter, AMIA Annual Symposium 2022 (N2C2 clinical NLP shared task)

PUBLICATIONS

Note: 'Kanjirangatt, V.' and 'Vani, K.' refer to the same author. Earlier publications use the surname-first format 'Vani, K.'

Conference & Workshop Papers

2026

1. **Kanjirangatt, V.**, Kletz, D., Samardžić, T., Dolamic, L. & Rinaldi, F. (2026). Evaluating Multilingual Tokenization under Worst-N Parity-Aware BPE. MeLLM Workshop, ACL 2026, pp. 221–228.
2. Mohan, V.M., Jayakumar, S., Menon, G.S., Kurup, S. & **Kanjirangatt, V.** (2026). When Retrieval Hurts: Evidence Utilization, Script Fidelity, and Knowledge Conflicts in Multilingual RAG. MeLLM Workshop, ACL 2026, pp. 92–107.
3. Menon, G.S., Jayakumar, S., Mohan, V.M., Kurup, S., Veena, G. & **Kanjirangatt, V.** (2026). Scope Matters: Evaluating Retrieval Strategies in Multilingual RAG Across Languages and Models. CLEF 2026. **[Accepted]**

2025

4. **Kanjirangat, V.**, Samardžić, T., Dolamic, L. & Rinaldi, F. (2025). Tokenization and Representation Biases in Multilingual Models on Dialectal NLP Tasks. EMNLP 2025 (Main Track), Suzhou, China. **SAC Highlight Award**
5. Lithgow-Serrano, O.*, **Kanjirangat, V.*** & Antonucci, A. (2025). Causal Understanding by LLMs: The Role of Uncertainty. UncertainNLP 2025, ACL Workshop, pp. 208–228. (*equal contribution)
6. Lithgow-Serrano, O., Kletz, D., **Kanjirangat, V.** et al. (2025). Assessing RAG System Capabilities on Financial Documents. FinNLP@EMNLP 2025, pp. 124–147.
7. **Kanjirangat, V.** & Rinaldi, F. (2025). Zero-Shot Cross-Sentential Scientific Relation Extraction via Entity-Guided Summarization. WASP@IJCNLP-AAACL 2025.
8. Sayeed, M.A., Gupta, D. & **Kanjirangat, V.** (2025). Engineering Text-to-text Generation LMs as Discriminative Classifiers. Procedia Computer Science, 258, 2930–2947.

2024

9. **Kanjirangat, V.**, Samardžić, T., Dolamic, L. & Rinaldi, F. (2024). NLP_DI at NADI 2024: Multi-label Arabic Dialect Classifications with an Unsupervised Cross-Encoder. ArabicNLP 2024 (ACL), pp. 742–747.
10. **Kanjirangat, V.**, Antonucci, A. & Zaffalon, M. (2024). On the Limitations of Zero-Shot Classification of Causal Relations by LLMs. CEUR-WS 2024.
11. **Kanjirangat, V.**, Dolamic, L. & Rinaldi, F. (2024). Dialect Identifications with Large Language Models. SwissText 2024.
12. Sayeed, M.A., Gupta, D. & **Kanjirangat, V.** (2024). Auto-Grading Comprehension using Siamese-based Transformer. IEEE I2CT 2024.

2023

13. **Kanjirangat, V.**, Samardžić, T., Dolamic, L. & Rinaldi, F. (2023). Optimizing the Size of Subword Vocabularies in Dialect Classification. VarDial 2023 (EACL), pp. 14–30.
14. **Kanjirangat, V.** & Antonucci, A. (2023). Edge Labelling in Narrative Knowledge Graphs. Text2Story 2023, ECIR 2023, Dublin.
15. Nair, P.C., Gupta, D., Devi, B.I. & **Kanjirangat, V.** (2023). Explainable Diagnostic Classification for Brain Tumor using Discharge Summaries. Procedia Computer Science, 218, 2058–2070.

2022

16. **Kanjirangat, V.**, Samardžić, T., Dolamic, L. & Rinaldi, F. (2022). Early Guessing for Dialect Identification. Findings of ACL: EMNLP 2022, pp. 6417–6426.
17. **Kanjirangat, V.**, Samardžić, T., Rinaldi, F. & Dolamic, L. (2022). NLP_DI at NADI Shared Task: Sub-word Level CNN Models for Dialect Identification. WANLP@EMNLP 2022, pp. 468–473.
18. Mitrović, S.* & **Kanjirangat, V.*** (2022). Enhancing BERT with Contextual Valence Shifters for Panic Detection in COVID-19 Tweets. NLPiR 2022. (*equal contribution)
19. Veena, G., Gupta, D. & **Kanjirangat, V.** (2022). Semi Supervised Approach for Relation Extraction in Agriculture Documents. OCIT 2022 (IEEE), pp. 199–204.

2019–2021

20. Mellace, S., **Vani, K.** & Antonucci, A. (2021). Relation Clustering in Narrative Knowledge Graphs. AI4Narratives@IJCAI-PRICAI 2021.
21. **Kanjirangat, V.**, Mitrovic, S., Antonucci, A. & Rinaldi, F. (2020). SST-BERT at SemEval-2020 Task 1: Semantic Shift Tracing by Clustering in BERT-based Embedding Spaces. SemEval@EMNLP 2020, pp. 214–221.
22. **Vani, K.**, Mellace, S. & Antonucci, A. (2020). Temporal Embeddings and Transformer Models for Narrative Text Understanding. Text2Story@ECIR 2020, pp. 71–77. **Best Paper Award**
23. Volpetti, C., **Vani, K.** & Antonucci, A. (2020). Temporal Word Embeddings for Narrative Understanding. ICMLC 2020 (ACM), pp. 68–72. **Best Paper Award**
24. Oita, M.* & **Vani, K.*** (2019). Semantically Corroborating Neural Attention for Biomedical Question Answering. ECML-PKDD 2019, pp. 670–685. (*equal contribution)
25. **Vani, K.** & Antonucci, A. (2019). NOVEL2GRAPH: Visual Summaries of Narrative Text Enhanced by Machine Learning. Text2Story@ECIR 2019, pp. 29–37.

2014–2016

26. **Vani, K.** & Gupta, D. (2016). ASE@DPIL-FIRE2016: Hindi Paraphrase Detection using NLP & Semantic Similarity. FIRE 2016, pp. 244–249.
27. Riya Ravi, N., **Vani, K.** & Gupta, D. (2016). Fuzzy C-Means Clustering in External Plagiarism Detection. ISTA 2016, Springer.
28. **Vani, K.** & Gupta, D. (2015). Combined Similarity Metrics and POS Tagging in Extrinsic Text Plagiarism Detection. ICACCI 2015 (IEEE), pp. 1578–1584.
29. **Vani, K.** & Gupta, D. (2014). K-Means Cluster Based Techniques in External Plagiarism Detection. IC3I 2014 (IEEE), pp. 1268–1273.
30. Gupta, D., **Vani, K.** & Singh, C.K. (2014). NLP and Fuzzy-Semantic Similarity for Automatic External Plagiarism Detection. ICACCI 2014 (IEEE), pp. 2694–2699.

Peer-Reviewed Journal Articles

2024

31. Nair, P.C., Gupta, D., Devi, B.I., **Kanjirangat, V.** & Deepak, P. (2024). Predicting Tumor Type and Residual Status of Suprasellar Lesions using Indian Discharge Summaries. IEEE Access.

2023

32. Veena, G., **Kanjirangat, V.** & Gupta, D. (2023). AGRONER: An unsupervised agriculture named entity recognition using weighted distributional semantic model. Expert Systems with Applications, 120440.
33. Veena, G., Gupta, D. & **Kanjirangat, V.** (2023). Semi-supervised Bootstrapped Syntax-Semantics based Approach for Agriculture Relation Extraction for Knowledge Graph Creation and Reasoning. IEEE Access.

2021

34. **Kanjirangat, V.** & Rinaldi, F. (2021). Enhancing Biomedical Relation Extraction with Transformer Models using Shortest Dependency Path Features and Triplet Information. Journal of Biomedical Informatics, 122, 103893.
35. Lithgow Serrano, O., Cornelius, J., **Kanjirangat, V.**, Méndez-Cruz, C.F. & Rinaldi, F. (2021). Improving classification of low-resource COVID-19 literature using Named Entity Recognition. Genomics & Informatics, 19(3).

2016–2018

36. **Vani, K.** & Gupta, D. (2018). Integrating syntax-semantic-based text analysis with structural and citation information for scientific plagiarism detection. JASIST, 69(11), 1330–1345.
37. **Vani, K.** & Gupta, D. (2018). Unmasking text plagiarism using syntactic-semantic based NLP techniques. Information Processing & Management, 54(3), 408–432.
38. **Vani, K.** & Gupta, D. (2017). Detection of idea plagiarism using syntax-semantic concept extractions with genetic algorithm. Expert Systems with Applications, 73, 11–26.
39. **Vani, K.** & Gupta, D. (2017). Text plagiarism classification using syntax-based linguistic features. Expert Systems with Applications, 88, 448–464.
40. **Vani, K.** & Gupta, D. (2017). Identifying document-level text plagiarism: A two-phase approach. Journal of Engineering Science and Technology, 12(12), 3226–3250.
41. **Vani, K.** & Gupta, D. (2016). Study on Extrinsic Text Plagiarism Detection Techniques and Tools. Journal of Engineering Science & Technology Review, 9(5).
42. Gupta, D., **Vani, K.** & Leema, L.M. (2016). Plagiarism detection using sentence bounded stop word n-grams. Journal of Engineering Science and Technology, 11(10), 1403–1420.

TECHNICAL SKILLS

Languages: Python, C++, PyTorch, TensorFlow, Keras, Scikit-learn

LLM & NLP: HuggingFace Transformers, SpaCy, PEFT (LoRA, Prefix-tuning, Prompt-tuning, P-tuning)

RAG & KGs: RAG pipeline design and evaluation

Evaluation & Responsible AI: ECE/ACE calibration, uncertainty quantification, bias auditing, hallucination evaluation

HPC & Infrastructure: CSCS (Swiss National Supercomputing Centre), GPU clusters, distributed LLM training, Linux

Domains: Arabic/Multilingual NLP, Biomedical NLP, Financial AI, ESG Analysis, Trustworthy AI, Educational NLP

CONTACT & PROFILES

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